One Auditable VLA Across Four Robot Suites A Fully Tensor-Decomposable Policy

Protocol-aligned control and exact weight analysis

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Abstract

Tensor decomposition turns a large table of learned interactions into simpler low-rank factors that can be ranked and inspected. Conventional robot policies contain operations that block this direct weight-level view. We introduce χ-VLA, a rational vision-language-action model family with a 19.2M convolutional policy and a 20.1M ViT policy, both built only from bilinear, linear, and rational learned operations. Their weights can therefore be reorganized into explicit interaction tensors rather than treated only as an opaque checkpoint. A runtime operator audit of the seed-0 20.1M ViT checkpoint and local numerical reconstructions verify its operator-level form. Across three independently trained 20.1M ViT policies, closed-loop LIBERO-Object success is $85.3\% \pm 4.6\%$ over 500 canonical trials per checkpoint. A fixed elementwise prediction-mean ensemble reaches $467/500 = 93.4\%$ at the cost of three forward passes. Separately, three jointly trained checkpoints spanning 40 familiar LIBERO tasks reach $70.35\% \pm 1.32\%$ task-macro success, compared with $69.12\% \pm 2.33\%$ for parameter-matched conventional controls under the same recipe. Their frozen three-pass mean-prediction ensemble reaches 74.75% with every suite at or above 50%. At equal aggregate updates, a frozen seed-0 joint checkpoint retains 95.16% of four suite specialists' pooled success. On the three LIBERO-Object checkpoints, a prospectively frozen paired test on 100 unique familiar-task states finds $247/300 = 82.3\%$ correct-instruction success, versus $72/300 = 24.0\%$ with a co-present control instruction and $78/300 = 26.0\%$ with an empty instruction. On disjoint episodes, zeroing only the learned post-lookup token vectors drops success from $263/300 = 87.7\%$ to $63/300 = 21.0\%$, with positive gaps for all ten task aggregates. A separate convolutional instantiation passes independent 4,608-case joint-attention, 384-case joint-FFN, and 384-case complete joint-block certificates. Their respective worst relative errors are 6.36×10^{-16}, 1.39×10^{-15}, and 1.5024×10^{-7}. Exact-attention records naming those same ViT checkpoint artifacts numerically replay every attention module on one fixed corrected cached input per checkpoint, with worst relative error 2.016119×10^{-7}. Across 48 fixed inputs, a sequential replay rebuilds all 12 learned transformer blocks to the action output at worst relative error 3.9822×10^{-7} against a frozen 10^{-3} gate. In a seed-0 ViT checkpoint, exact weight-derived subspaces beat equal-rank random controls in all 36 primary block-head tests. Separately, prospectively fixed rank-96 visual projectors retain 95.2–98.2% of baseline successes on tasks absent from corrected projector construction, while three equal-rank random projectors retain only 3.3–13.6%. The result is a capable and structurally inspectable single-suite and four-suite familiar-task family with exact access to trained attention coefficients and bounded familiar-task instruction

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necessity. It does not establish counterfactual goal following, zero-shot language use, or an input-general compact symbolic decomposition of the full policy.

1 Introduction

Modern robot policies can achieve high success while giving little indication of why an action was chosen. Their checkpoints contain millions of weights, but those weights do not directly show which image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated activations, yet a correlation is not automatically a mechanism or a safe editing target.

This is not a zero-shot policy. Every evaluated task appears during training. We test both single-suite specialists and one jointly trained four-suite policy to establish auditable familiar-task control against which future zero-shot and transfer claims can be evaluated.

A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction table defined by its weights. Its decomposed factors then provide candidate control mechanisms that can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully tensor-decomposable policy. It is an operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations.

The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization break the algebraic form, while continuous actions require precise closed-loop outputs. The potential payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows language, keys on a visual shortcut, or uses a fragile control feature.

We study the narrower but testable question:

> *Can a compact VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?*

We test capability and structural analysis in two fully rational encoder instantiations. The same three ViT checkpoints establish closed-loop capability and support the exact attention audit. A separate convolutional instantiation tests whether structural certificates transfer across visual encoders.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate three fully rational 20.1M ViT policies on $1{,}500$ canonical LIBERO-Object trials. They reach $85.3\% \pm 4.6\%$, while a fixed prediction-mean ensemble reaches $467/500 = 93.4\%$. Three jointly trained checkpoints average $70.35\% \pm 1.32\%$ over 40 familiar tasks, 1.23 points above matched conventional controls. Their fixed ensemble reaches 74.75%. A frozen seed-0 comparison retains 95.16% of four suite specialists.
3. We mechanically audit the seed-0 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor. In a separate closed-loop audit, learned rank-96 visual projectors retain 95.2–98.2% of baseline successes, versus 3.3–13.6% for random projectors.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive replaces a familiar transformer component while keeping explicit coefficients. Linear maps contribute first-order terms, while multiplying learned projections creates higher-order interactions that remain determined by the weights.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho}L_{\rho i}R_{\rho j}\right)\bar{x}_i\bar{x}_j. \tag{1}$$

The inner sum is a table indexed by one output and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1K_1^\top) \odot (Q_2K_2^\top)/s\right]V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with interaction coefficients determined by the weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. Let $m(x) = d^{-1}\sum_{j=1}^{d} x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \qquad \mathrm{RNorm}(x) = \frac{x}{\sqrt{s_0}}\,r(v). \tag{3}$$

Here rational means a ratio of polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D. Updating this pair preserves input-specific rescaling without leaving the rational tensor representation. Training and deployment use r itself. The exactness claims below therefore concern faithful evaluation of the deployed rational computation, not uniform agreement with RMSNorm. Appendix A.5 certifies pole-free denominators and deployed arithmetic fidelity while retaining that boundary.

2.2 Architecture and training

Both encoder variants follow the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both share a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-query state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision blocks and has 20,137,352 learned parameters. It supplies the closed-loop capability, runtime operator census, exact-attention, and reduced-Gram results. The 19.2M convolutional variant instead uses three stride-2 bilinear convolution stages that produce an 8×8 token grid. Three separate convolutional files receive the joint-attention and rational-normalizer certificates reported in Appendix A.5.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the seed-0 20.1M ViT checkpoint

Architecture definitions can differ from deployed code. We therefore audit the deployed seed-0 20.1M ViT checkpoint at runtime and reconstruct representative operations from saved weights.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction, as summarized in Appendix Figure 6. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with the following fixed protocol:

- 50 canonical trials per task
- a 280-step cap
- three independently trained ViT checkpoints
- 500 rollouts per checkpoint and 1,500 rollouts in total

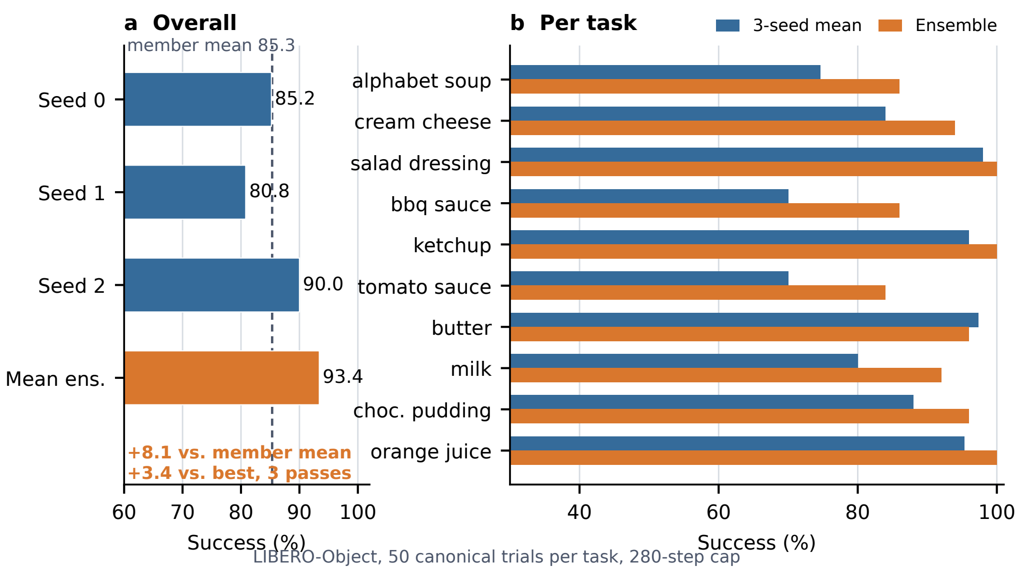

Figure 1: **Verified ViT capability.** Left: three independently trained checkpoints and their fixed elementwise prediction-mean ensemble. Right: per-task three-checkpoint mean and ensemble success. The ensemble uses three forward passes.

3.2 Results

Method	Parameters	Inference	Success
χ-VLA, rational ViT	20.1M	1 pass	$\mathbf{85.3\% \pm 4.6\%}$
χ-VLA, prediction-mean ensemble	3×20.1M	3 passes	$\mathbf{93.4\%}$

Table 1: **Closed-loop LIBERO-Object success.** Each ViT checkpoint receives 500 canonical trials. The single-model row is the mean $\pm$ sample standard deviation across three checkpoints. The ensemble is one fixed composite evaluated on the same 500 states.

The three checkpoints reach $426/500 = 85.2\%$, $404/500 = 80.8\%$, and $450/500 = 90.0\%$. Elementwise averaging of their predicted action chunks reaches $467/500 = 93.4\%$. This is 8.1 points above the member mean and 3.4 points above the strongest member. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy. Every episode-level record, task index, and protocol field is retained. The artifact manifest binds each immutable result file and recorded checkpoint basename to a checkpoint hash verified separately. Member and ensemble shards span different recorded GPU families and predate explicit matrix-precision fields, so the arithmetic gaps are descriptive rather than a matched-hardware ensemble ablation.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

3.4 One policy across four familiar suites

Three jointly trained 20.1M checkpoints reach $70.35\% \pm 1.32\%$ task-macro success over 40 familiar LIBERO tasks and 50 trials per task, 1.23 points above parameter-matched conventional controls at $69.12\% \pm 2.33\%$ under the same recipe. Their frozen three-pass mean-prediction ensemble reaches 74.75%, with every suite and $34/40$ tasks at or above 50%. At equal aggregate updates, a frozen seed-0 joint checkpoint retains 95.16% of four suite specialists, above the fixed 85% gate. Because all 40 tasks appeared in training, this establishes multi-suite capacity, not unseen-task transfer.

3.5 Familiar-task instruction necessity

Before observing these outcomes, we fixed a paired full-horizon test on 100 unique canonical states across the same three checkpoints. Correct instructions succeed on $247/300 = 82.3\%$ of checkpoint-state pairs, compared with $72/300 = 24.0\%$ for an observation-verified co-present control instruction and $78/300 = 26.0\%$ for an empty instruction. Both gaps are positive for every checkpoint aggregate and every task aggregate. The exact tests and a state-cluster bootstrap also pass their frozen gates. Appendix A.1 gives the protocol, statistics, and plot. Every tested task and instruction appeared during training, so this is instruction necessity for familiar tasks, not zero-shot language generalization or counterfactual goal following.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. On one fixed corrected cached input per checkpoint, we numerically replay all four vision and eight joint attention modules in each of three records that name the same ViT checkpoint artifacts as the capability results. This covers 36 module-input cases and 384 architectural heads. The worst relative ℓ_2 error is 2.016119×10^{-7} and the worst absolute difference is 2.44×10^{-5}. This residual comes from the deployed implementation's intentional float32 statistic, while the reconstruction itself uses the learned coefficients. The algebraic identity is input-general. The one-input-per-checkpoint records provide a layerwise numerical replay audit, not an exact end-to-end policy decomposition.

We separately certify three convolutional checkpoints from the same rational architecture family. Independent reconstruction covers $4{,}608$ joint-attention head-input cases, 384 joint-FFN module-input cases, and 384 complete joint-block input cases. Their respective worst relative errors are

6.36×10^{-16}, 1.3871×10^{-15}, and 1.5024×10^{-7}, all below their frozen gates. These are input-conditioned layer and block certificates.

A frozen ViT audit partitions 36,864 joint-attention head-input cases into vision, instruction, robot-state, and action-query sources. Worst reconstruction error is 5.0741×10^{-7}, while mean coherent-energy shares are 67.7%, 5.1%, 19.2%, and 7.9%. The shares are descriptive rather than causal importance or grounding.

On the same checkpoints, a frozen sequential replay rebuilds the learned path from prepared tensors through all 12 transformer blocks to the linear head. Over 48 inputs and 816 stages, worst final-action relative error is 3.9822×10^{-7} against a frozen 10^{-3} gate. This fixed-input numerical certificate is not a compact symbolic or input-general proof.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors rank important directions without constructing the full table. It matches brute-force materialization to 2.13×10^{-14} at toy scale.

4.2 Trained policy result

The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached activations are used only afterward to score how well it preserves the layer's computation.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of 36 block-head pairs favor the exact subspace, with median ratios 1.028, 1.171, and 2.441. At rank 128, the mean is 3.033 and the maximum is 10.135. These are architectural units within one checkpoint, not independent model-level replicates.

This extends exact weight-only construction through individual attention layers. It is not an exact compact symbolic end-to-end decomposition of the complete policy. Symbolic composition across all eight blocks still faces rapid degree and representation growth.

A rank-96 action-Jacobian projector fixed from official Object tasks 0–3 retains $259/272$ full-policy successes on tasks 8–9 across three ViT checkpoints. An activation-energy projector retains $267/272$, while three equal-rank random projectors retain $37/272$, $9/272$, and $22/272$. All ten tasks appeared during policy training, but tasks 8–9 are absent from corrected projector construction. This supports robustness to a structured 75% channel reduction, not zero-shot transfer, unique action-Jacobian specificity, or a causal link to the exact weight factors. Appendix A.4 reports paired controls and stability intervals.

5 Related work and limitations

Related work. OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control baselines [1, 2]. Bilinear MLPs, χ-nets, and bilinear IMPALA policies show that multiplicative weights can expose low-rank structure and support subsequent controls [3–5]. Recent work also steers pretrained VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a policy designed as a rational tensor program. Tensor and polynomial networks provide the broader multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success can coexist with weak language use [9–11].

Limitations and conclusion. The breadth benchmark covers 40 familiar training tasks across four suites. Exact layer equations and a fixed-input learned-forward replay reach the action output, but neither is a compact symbolic or input-general proof. The paired audit establishes familiar-task instruction necessity, not counterfactual goal following, zero-shot language use, or broad grounding. Within that scope, rational restrictions support strong control and trained-weight certificates. We do not claim unseen-task generalization or coefficient editing.

References

[1] M. J. Kim et al. *OpenVLA: An Open-Source Vision-Language-Action Model.* CoRL, 2024.

[2] M. J. Kim, C. Finn, and P. Liang. *Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success.* 2025.

[3] A. Rigg, T. Dooms, M. T. Pearce, J. Oramas, and L. Sharkey. *Bilinear MLPs Enable Weight-Based Mechanistic Interpretability.* ICLR, 2025.

[4] T. Dooms, W. Gauderis, G. A. Wiggins, and J. Oramas. *Compositionality Unlocks Deep Interpretable Models.* 2025.

[5] F. Oozeer, Y. Erisken, A. Rigg, J. Oramas, and L. Sharkey. *Bilinear Convolution Decomposition for Causal RL Interpretability.* 2024.

[6] B. Häon, K. Stocking, I. Chuang, and C. Tomlin. *Mechanistic Interpretability for Steering Vision-Language-Action Models.* 2025.

[7] S. Marks et al. *Sparse Feature Circuits: Discovering and Editing Interpretable Causal Graphs in Language Models.* ICLR, 2025.

[8] M. Kekić et al. *Learning Nonlinear Causal Reductions to Explain Reinforcement Learning Policies.* ICLR, 2026.

[9] X. Zhou et al. *LIBERO-PRO: Towards Robust and Fair Evaluation of Vision-Language-Action Models Beyond Memorization.* 2025.

[10] Y. Fang et al. *When Vision Overrides Language: Evaluating and Mitigating Counterfactual Failures in VLAs.* 2026.

[11] C. Kim et al. *LIBERO-Para: A Diagnostic Benchmark and Metrics for Paraphrase Robustness in VLA Models.* 2026.

[12] T. Kolda and B. Bader. *Tensor Decompositions and Applications.* SIAM Review, 2009.

[13] E. Stoudenmire and D. Schwab. *Supervised Learning with Tensor Networks.* NeurIPS, 2016.

[14] G. Chrysos et al. *Deep Polynomial Neural Networks.* CVPR and TPAMI, 2020 to 2021.

A Supporting results and decisions

A.1 Prospectively frozen closed-loop instruction necessity

High task success alone does not show that a policy uses its instruction. Before these outcomes were observed, we fixed a paired test on canonical episodes 40–49 for every task and the same three checkpoints. Each checkpoint-state pair was rolled out for the full 280-step horizon under the correct familiar instruction, one deterministic outcome-independent co-present control instruction, and an empty instruction. All conditions shared the exact initial physical input and their order was chosen by a frozen hash. Co-presence was verified from simulator observation fields and does not guarantee unoccluded camera visibility. The empty string encodes tokenizer BOS plus 31 unmasked PAD tokens, together with the model's learned common BOS.

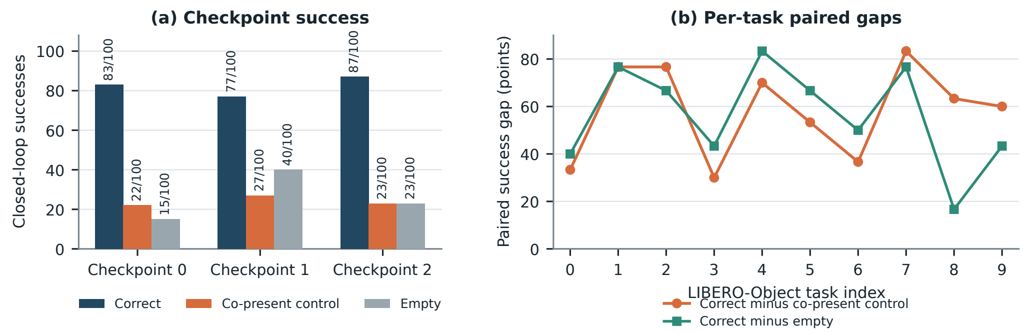

Figure 2: **Closed-loop instruction necessity on familiar tasks.** Left: exact successes among 100 paired canonical states per checkpoint. Right: correct-instruction success minus each control, pooling the three checkpoint outcomes for each task. All ten task gaps are positive for both controls at the task-aggregate level.

Across 100 unique canonical states and three checkpoints, the correct instruction succeeds on $247/300 = 82.3\%$ of checkpoint-state pairs. The co-present control succeeds on $72/300 = 24.0\%$ and the empty instruction on $78/300 = 26.0\%$, giving paired gaps of 58.3 and 56.3 points. Correct-instruction success is $83/100$, $77/100$, and $87/100$ by checkpoint, and both paired gaps are positive for every checkpoint aggregate. All ten pooled task aggregates are strictly positive for both controls. One-sided exact McNemar tests give $p = 2.02 \times 10^{-45}$ and $p = 1.52 \times 10^{-44}$. These pooled tests use 300 checkpoint-state pairs and are not cluster-robust inference. A frozen 20,000-draw task-stratified state-cluster bootstrap resamples 100 unique task-episode states while retaining all three checkpoint outcomes. Its 95% intervals are $[51.0, 65.7]$ and $[50.0, 62.3]$ points. Every prospectively frozen gate passes.

This establishes instruction necessity for familiar LIBERO-Object prompts, objects, scenes, tasks, and checkpoints. Appendix A.2 gives a disjoint paired test that isolates the learned lexical-embedding path. Neither test establishes counterfactual goal following, paraphrase robustness, unseen-object or compositional grounding, cross-suite or physical transfer, or a benefit unique to tensor decomposability.

A.2 Causal isolation of the learned instruction-embedding path

Token IDs are lookup addresses. A learned embedding table turns those addresses into vectors that the policy can combine with vision and robot state. Before observing outcomes, we fixed a paired test that isolates this conversion. On canonical episodes 30–39, disjoint from the instruction-necessity episodes, both arms received the identical correct 32-token ID tensor. The ablated arm replaced only the output of the token-embedding lookup with exact zeros. Sequence length, token positions, vision, robot state, embodiment, the separate learned common BOS, action queries, transformer weights, and action head were preserved and hash-checked.

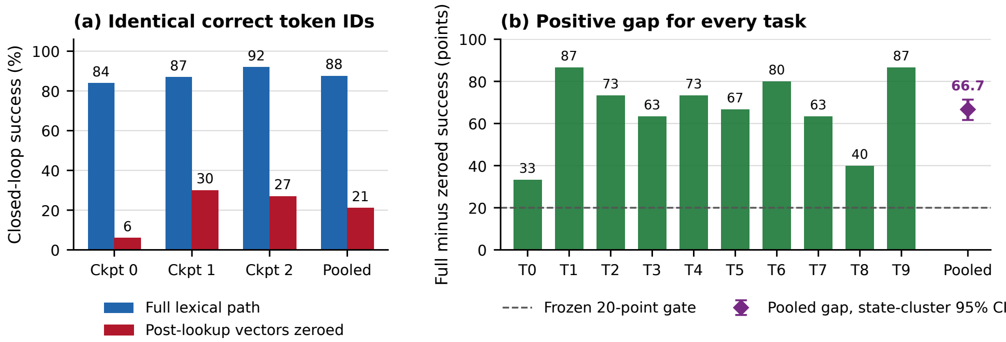

Figure 3: **Learned lexical-embedding-path necessity on familiar tasks.** Left: closed-loop success for the intact path and exact post-lookup zeroing, using 100 paired states per checkpoint and 300 pooled pairs. Right: every task aggregate has a positive paired gap. The pooled marker shows the frozen task-stratified state-cluster 95% bootstrap interval.

The intact path succeeds on $263/300 = 87.7\%$ of checkpoint-state pairs, compared with $63/300 = 21.0\%$ after zeroing the learned token vectors. The paired gap is 66.7 points, with 203 intact-only and 3 zeroed-only outcomes. Full-path success is 84%, 87%, and 92% by checkpoint, and the corresponding gaps are 78, 57, and 65 points. All ten task aggregates are positive. A pooled one-sided exact paired test gives $p = 1.42 \times 10^{-56}$, but this test is not cluster-robust because checkpoints reuse the same 100 states. The frozen 20,000-draw task-stratified state-cluster bootstrap interval is $[61.7, 71.3]$ points and is our cluster-aware inference. All eight prospectively fixed gates pass.

This establishes that the learned lexical-embedding path is necessary for these familiar LIBERO-Object tasks and three fixed policies. It does not establish semantic grounding, paraphrase robustness, unseen-object or compositional generalization, cross-suite or physical transfer, or a benefit unique to tensor decomposability.

For WRL, the specialist uses familiar task language, but every tested task and instruction appeared during training. The result is therefore not zero-shot language generalization.

A.3 Expanded attention screen

Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.4 Prospectively fixed-rank visual bottleneck

The behavioral intervention acts at the width-384 visual-to-joint interface and keeps rank 96. Projector construction uses official Object tasks 0–3. Rank 96 was fixed before any corrected-basis outcome was observed, and the frozen evaluation uses tasks 8–9. Across three checkpoints and 300 canonical episodes, the full policy succeeds on 272 trials. The action-Jacobian projector retains $259/272 = 95.2\%$ of those successes, with a paired checkpoint-task-episode resampling interval of 91.2–98.6%. The activation-energy projector retains $267/272 = 98.2\%$, with an interval of 95.8–100.0%. Three fixed equal-rank random projectors retain $37/272 = 13.6\%$, $9/272 = 3.3\%$, and $22/272 = 8.1\%$.

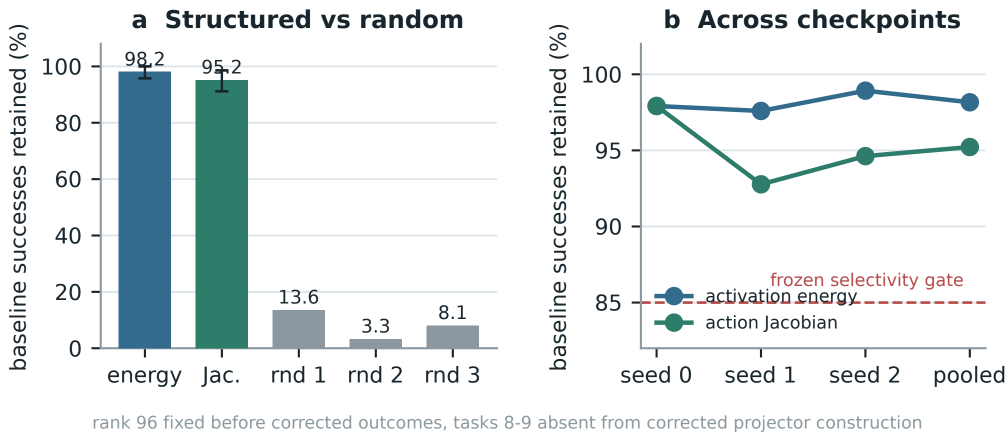

Figure 4: **Corrected closed-loop visual bottleneck.** Both learned rank-96 projectors preserve nearly all baseline successes across three checkpoints, while equal-rank random projectors do not. Error bars in panel (a) are deterministic paired cluster-resampling intervals. Tasks 8–9 are absent from corrected projector construction, but not from policy training. This is not a clean rank-selection-holdout result.

The action-Jacobian basis does not outperform the matched activation-energy basis. The stronger action-specificity gate therefore does not pass. The supported claim is structured visual sufficiency relative to random projectors. This is not zero-shot evidence, and the data-derived intervention remains separate from the exact weight-only attention decomposition.

A.5 Numerical scope of rational conversion

The deployed degree-$[2/2]$ least-squares rational fit is exact as a rational computation even though it approximates RMS normalization. Its numerator has coefficients $(5.32997, 7.53552, 0.327274)$ and the denominator has coefficients $(1, 9.64827, 2.60664)$. Every denominator coefficient is positive, which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances, each invoked once per forward pass, on $4{,}096$ deterministic inputs for each of three convolutional checkpoints. Across $544{,}542{,}720$ activation rows, the minimum observed denominator is 1.0016 and no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3} gate. This primary certificate concerns pole-free denominators and floating-point fidelity to the deployed rational operator. It does not claim uniform agreement with exact RMS scaling or measure matched alternative-normalizer runtime overhead.

The reconstruction ladder extends from individual attention heads through FFNs and per-block replay to a separate sequential learned-forward certificate. That final record starts at exact prepared model tensors and reaches the linear action head on 48 fixed inputs. It is numerical and input-conditioned, not a compact symbolic contraction or an input-general full-policy proof. The ViT source-partition audit remains layerwise and descriptive. Figure 5 reports the frozen numerical gates and the depth profile without assigning causal importance.

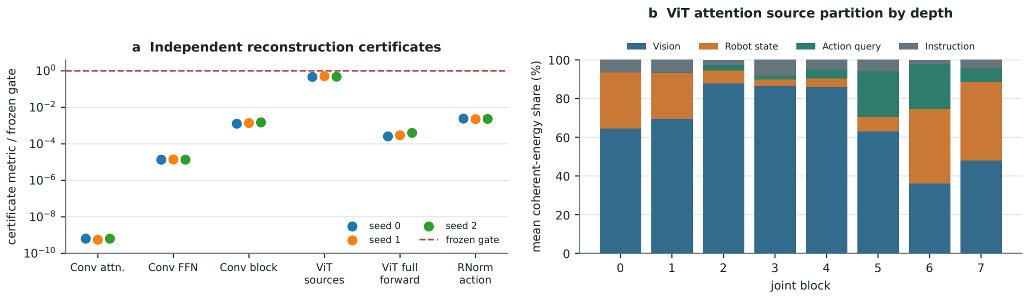

Figure 5: **Frozen reconstruction ladder and descriptive ViT source partition.** Panel (a) reports observed-to-gate ratios for independent Conv attention, joint-FFN, and complete joint-block reconstruction, the ViT source-partition identity, the sequential ViT learned-forward action replay, and primary RationalNorm action fidelity. All plotted primary numerical gates pass. Panel (b) shows descriptive mean coherent-energy shares after the attention output projection across 384 checkpoint-input cases per joint block. The depth trend is not a causal importance or grounding result. The learned-forward replay excludes preprocessing, fixed action unnormalization, chunk execution, simulator dynamics, and success.

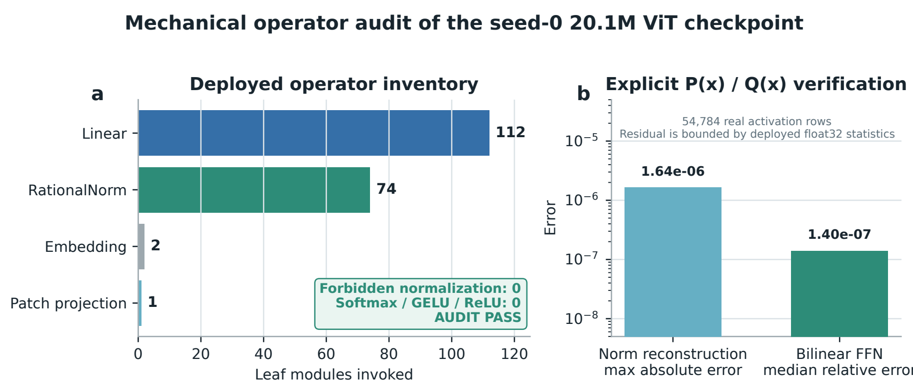

Figure 6: **Mechanical audit of the seed-0 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over $54{,}784$ real activation rows.